

Data-Driven Inventory Analysis: Mint Classics Case Study

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Objective: Evaluate current warehouse inventory levels, examine sales patterns, identify underperforming products, and provide strategic recommendations on closing one storage facility while preserving the company's 24-hour shipping commitment.

1. Executive Summary & Project Scenario

The Mint Classics Company, a retailer of classic model vehicles, is assessing options to optimize its logistics network by closing one of its four storage facilities. This business decision must be heavily data-driven to eliminate unnecessary warehousing costs without introducing operational bottlenecks or degrading customer service (specifically the mandate to ship orders within 24 hours).

Through Exploratory Data Analysis (EDA) using MySQL, we analyzed current inventory levels against historical sales metrics. The dataset reveals clear structural inefficiencies—primarily vast amounts of overstocking across almost all product lines and dead stock that binds warehouse space without turning a profit.

2. Key Explanatory Data Insights

A. Storage Facility Optimization (Warehouse Evaluation)

An aggregate query evaluated total products and active stock distribution across the four facilities:

- **Warehouse b (East):** Operates as the main hub with 38 product lines and 219,183 items.
- **Warehouse a (North) & c (West):** Maintain moderate inventory levels with 131,688 and 124,880 items respectively.
- **Warehouse d (South):** Holds the lowest inventory footprint, housing only 23 product lines and a total of 79,380 items.

Strategic Insight: Due to its significantly lower volume, Warehouse d (South) is the primary candidate for decommissioning.

B. Chronic Overstocking Analysis

The core finding of this analysis is an extreme misalignment between supply and demand. Multiple high-volume SKUs maintain thousands of units in stock while historical order tallies are minimal. For instance, the *2002 Suzuki XREO* in Warehouse a retains a current stock level of 9,997 units against a total lifetime sales volume of only 1,028 units. This indicates that existing stock levels are inflated beyond reasonable short-to-medium-term forecasting projections.

C. Identification of Dead Stock

Crucially, certain high-volume items exhibit no market traction whatsoever. Most notably, the **1985 Toyota Supra** stored in Warehouse b holds **7,733 units in stock** with a total sold metric of NULL (zero historical orders). This represents severe asset immobilization.

3. Final Data-Driven Recommendations

1. **Decommission Warehouse d (South):** Consolidate its 79,380 items into Warehouses a, b, and c.
2. **Inventory Downsizing & Right-Sizing:** Reduce the baseline stock numbers of heavily overstocked items (such as the Suzuki XREO) by 40% to 50% upon the next operational cycle, freeing massive physical space across remaining facilities.
3. **Product Line Pruning (Liquidation):** Immediately discontinue and liquidate dead stock items—specifically the 1985 Toyota Supra—to instantly recover lost shelf capacity.
4. **Preserving 24-Hour Fulfillment:** By modernizing replenishment schedules based on actual rolling sales velocities instead of static high-volume storage, the company can comfortably handle customer order fulfillment within the required 24-hour window from three optimal regional hubs.

4. Complete Project SQL Script

The following well-documented SQL script contains the full analytical exploration pipeline used to extract these business-critical insights from the `mintclassics` schema.

```
/*
=====
MINT CLASSICS INVENTORY & WAREHOUSE OPTIMIZATION SCRIPT
Author: Data Analyst
Purpose: Exploratory Data Analysis (EDA) for Facility Closure
=====
*/

-- Step 0: Initialize database context
USE mintclassics;

-- =====
-- QUERY 1: Warehouse Inventory Profiles
-- Objective: Identify total stock volume and product variety per facility.
-- =====
SELECT
    w.warehouseCode,
    w.warehouseName,
    COUNT(p.productCode) AS total_products,
    SUM(p.quantityInStock) AS total_stock
FROM warehouses w
LEFT JOIN products p ON w.warehouseCode = p.warehouseCode
GROUP BY w.warehouseCode, w.warehouseName
ORDER BY total_stock DESC;

-- =====
-- QUERY 2: Inventory to Sales Velocity Evaluation
```

```
-- Objective: Detect overstocking patterns by contrasting current stock
--           against lifetime historical sales volumes.
-- =====
SELECT
    p.productCode,
    p.productName,
    p.warehouseCode,
    p.quantityInStock AS current_stock,
    SUM(od.quantityOrdered) AS total_sold,
    (p.quantityInStock - SUM(od.quantityOrdered)) AS stock_vs_sales_difference
FROM products p
LEFT JOIN orderdetails od ON p.productCode = od.productCode
GROUP BY p.productCode, p.productName, p.warehouseCode, p.quantityInStock
ORDER BY stock_vs_sales_difference DESC;
```

```
-- =====
-- QUERY 3: Dead Stock and Candidate Products for Line Pruning
-- Objective: Locate products with substantial stock but zero sales.
-- =====
SELECT
    p.productCode,
    p.productName,
    p.warehouseCode,
    p.quantityInStock AS current_stock,
    SUM(od.quantityOrdered) AS total_sold
FROM products p
LEFT JOIN orderdetails od ON p.productCode = od.productCode
GROUP BY p.productCode, p.productName, p.warehouseCode, p.quantityInStock
HAVING total_sold IS NULL OR total_sold = 0
ORDER BY current_stock DESC;
```